

First Last, Ph.D.

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Experience

Senior Data Scientist

June 2022 - Present

Company name

- Developed a YOLO-based object-detection model in PyTorch that achieved a 80% mAP evaluation on a in-house thermal-image vehicle dataset.
- Designed an anomaly-detection model in PyTorch using an LSTM variational-autoencoder to detect component failure in a pulse-power x-ray machine. The model accurately predicted the location of a component failure with 87% accuracy and cut machine repair times by 50%.
- Applied the U-Net architecture with PyTorch to develop a deblurring model for a pulse-powered x-ray machine. The model was applied by the machine operators and reduced machine configuration time by 30%.
- Facilitated the scaling of locally developed models to be trained on our internal high-performance computing (HPC). Would take code written by the other Data Scientists and Analysts and would refactor so that they could run in a multi node, multi gpu setting.
- Developed interactive dashboards using Streamlit to give non-technical staff the ability to interact with the capabilities designed by our team of Data Scientists.
- Mentored a summer intern in the development and execution of models on a HPC cluster, involving multi-node, multi-GPU setups and SLURM resource management.
- Sole developer of a Django web interface, which wrapped Neo4j. The interface gave non-technical staff the ability to query and update the company's Neo4j graph-database without needing to know any query syntax.

Technical Skills

Programming Languages: Python, SQL, Bash, Cypher, Lua, L^AT_EX

Libraries: PyTorch, Scikit-Learn, Pandas, Scipy, Sqlalchemy, Matplotlib, Plotly, Streamlit, Django

Developer Tools: Git, AWS, Unix/Linux, Slurm, Vim, Tmux, Neo4j

Education

University 2 - Ph.D.

June 2022

Research Area: Probability (Stochastic Processes and Partial Differential Equations)

Advisor: First Last

University 1 - B.S./M.S.

December 2015 / January 2018

Major: Applied Mathematics

Publications

1: Q1 probability and stochastic process journal

2: Q2 probability journal